

Fake News Detection in Portuguese Under Large Language Model-Generated Content

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Abstract. We are daily exposed to fake news, a growing problem that spreads in various forms, including rumours, advertisements, social media posts, and political propaganda. Predominantly created by humans, in recent years, we have witnessed an increase of digital content fabricated or manipulated with the use of deep learning. Large Language Models (LLMs), for instance, represent a real threat if used to generate highly convincing fake news that could evade conventional detection systems. This study evaluates the impact of LLM-generated fake news on machine learning (ML) classifiers. The ML models are trained with Portuguese-language datasets and experiments are conducted using aligned data, where each fake news sample has its true news counterpart. We assess the performance of each ML model with synthetic fake news test set, which was generated by a Portuguese-specific LLM. Our results reveal significant performance degradation of ML models when evaluated on LLM-generated false information. These findings highlight the need for updated detection strategies capable of handling the linguistic and stylistic nuances introduced by LLMs. To address that, a Retrieval-Augmented Generation (RAG) framework was evaluated under the same conditions as the ML models. The framework showed to be more robust when there was a distribution shift between train and test data, whereas ML models provided better performance when the data remained consistent. These results contribute to the understanding of fake news detection in Portuguese, emphasizing the importance of adapting existing models to the evolving nature of misleading content.

Keywords: Fake news, Machine Learning, Large Language Models, Retrieval-Augmented Generation

1 Introduction

Communication has become more complex with the ubiquity of social networks [Aïmeur *et al.*, 2023]. The latter certainly offers a number of benefits to society, from an easy communication to family members living far away to the possibility of authorities sharing vital information during catastrophic natural disasters [Saroj and Pal, 2020]. Nevertheless, they also enables the easy production and rapid dissemination of fake news [Shu *et al.*, 2017]. During the COVID-19 pandemic, for example, while several governments strived to inform and protect the public, an overload of inaccurate information, also known as infodemics [Gallotti *et al.*, 2020], was being consumed on social media platforms [Smith *et al.*, 2021]. In such outbreaks, the spread of false information can put individuals' life at risk if they decide to undermine authorities' recommendations [Rathore and Farooq, 2020].

There are many factors that contribute to deficient reasoning while deciding what is true or false [Ecker *et al.*, 2022]. The great part of it is related to human psychology and social interaction with peers. While high quality information or knowledge about specific subjects are important, it is not

the primary factor driving misinformation beliefs [O'Connor and Weatherall, 2021]. Studies have indicated, for example, that people are more likely to share false information than truth [Vosoughi *et al.*, 2018], which can worsen the issues created by fake news, including preventing information aggregation and consensus within the population [Azzimonti and Fernandes, 2023]. Additionally, it is also found that individuals tend to believe on information learned from people that they trust as a way to minimize the need for acquiring domain-specific knowledge to form reliable beliefs [O'Connor and Weatherall, 2021].

The increasing sophistication of AI models, capable of generating high-quality deepfake signals (e.g., text, image, and audio), is yet a threat to information integrity in multi-media communication. Unlike the traditional news media (e.g., newspaper, TV, and radio) where published news is expected to be scrutinized and also associated to an author, social media has enabled the rapidly spread of misleading anonymous information, known as fake news, almost unrestrictedly. Here, we follow the same definition of fake news offered by Lazer *et al.* [2018], in which the term refers to fabricated information with no regard to accuracy and with

the lack of the journalistic standards and organizational process. Although often used interchangeably, it's important to note that fake news can refer to two types of misleading content: misinformation, which is defined as any information that happens to be false; and disinformation, which refers to the subset of misinformation intentionally spread with the intent to mislead.

The phenomenon of fake news has been studied for several years, and despite the increasing attention it has been receiving, several challenges still remain. For example, the lack of studies based on low-resource languages and the predominant focus on English are considered key limitations in existing research by Plikynas *et al.* [2025]. Thus, generalizing conclusions from English-based studies is not always evident nor applicable to languages with distinct characteristics, such as Portuguese. Plikynas *et al.* [2025] also highlight the challenge faced by traditional learning algorithms when confronted with the dynamic nature of fake news. This is in line with studies exploring the impact of data shift on machine learning (ML) models used for fake news detection [Silva *et al.*, 2023].

Additionally, in recent years, we have witnessed the rise and widespread adoption of Large Language Models (LLMs). GPT-4 (used in ChatGPT) [Achiam *et al.*, 2023], LLaMA [Touvron *et al.*, 2023], Sabiá-3 (used in MariTalk, a chatbot similar to ChatGPT) [Abonizio *et al.*, 2024], and DeepSeek-R1 [DeepSeek-AI, 2025], for instance, are potential assets for creating fake news. These models can ultimately enable anyone to generate highly convincing textual content. This is a real threat to ML models trained on fake news datasets primarily developed with human-written texts. As these models operate in a static manner, their performance on new data distributions is often degraded.

A study by Sardinha [2024] reveals that, despite AI-generated texts appearing convincingly human-like, a detailed linguistic analysis shows significant differences between human and AI-generated text. These differences indicate a distributional shift in textual features, raising the question of how to cope with new types of content produced by LLMs, which may not be effectively handled by traditional detection models. As the study by Sardinha [2024] was not focused on fake news nor on the Portuguese-Language, the present study aims to address this gap by investigating whether similar patterns are present in the Portuguese-Language used in such content.

We hypothesize that human and LLM-generated fake news exhibits distinct linguistic characteristics also in Portuguese. Moreover, we expect that the presence of such differences will affect the ability of traditional ML models to generalize to LLM-generated misleading information, as these models are typically trained on human-written fake news. However, the impact of these distinctions on the performance of ML methods remains an open question. We address this by investigating how model performance differs when dealing with LLM-generated versus original fake news. Hence, our study contributes to a clearer understanding of the challenges and limitations of traditional ML methods in detecting fake news in this evolving scenario.

To date, several approaches to detect human-generated fake news have been proposed [Silva *et al.*, 2020; Cabral

et al., 2021; Chavarro *et al.*, 2023; Garcia *et al.*, 2024]. One can ultimately choose between traditional ML solutions or more advanced models based on deep neural network (DNN). Despite the availability of such advanced techniques, more compact and simpler models is often desired for lowering financial and computational costs. To evaluate the simplicity of traditional ML models and the potential capabilities of advanced language models, we fine-tuned the Bidirectional Encoder Representations from Transformers (BERT). This enables us to compare the robustness of traditional ML models and DNNs.

Furthermore, to overcome the limitations of static models in traditional ML, we explored the use of LLMs combined with external knowledge. This can be achieved with the so-called Retrieval-Argumented Generation (RAG) framework [Lewis *et al.*, 2020a]. The comparison between traditional ML and RAG is particularly relevant because to the best of our knowledge, no prior study has explored RAG-based fake news detection in Portuguese. The closest work to this is Gôlo *et al.* [2024], which applied prompts directly to LLMs without leveraging the external knowledge available with RAG. Evaluating these methods not only helps assess the impact of synthetic fake news on detection performance but also determines whether this recent approach surpasses traditional ML methods.

In summary, our main contribution lies on the investigation of the impact of LLM-generated fake news on traditional ML methods and advanced approaches such as BERT and RAG, which to date, has not been explored in the context of Portuguese. Additionally, while some studies have explored LLM-generated fake news in English [Ayoobi *et al.*, 2024; Jiang *et al.*, 2024], none have analyzed their impact on automatic classification using aligned datasets — where each fake news sample has its true counterpart. We believe this approach leads to a more precise and reliable analysis by better isolating the differences between true and false news, ensuring that any variation in the model performance stems from the false information phenomenon itself rather than structural differences in the data. Moreover, comparing fake and true news about the same event helps reduce biases related to the topic, writing style, or linguistic complexity.

The remainder of this paper is structured as follows. Section 2 presents the main related work in the field. Section 3 details the use of an LLM for fake news generation. Section 4 details the materials and methods employed in the research. Section 5 covers the results. Finally, Section 6 concludes the paper and offers guidelines for future research.

2 Related Work

In this section, we provide a concise overview of the evolution of fake news detection, followed by the state-of-the-art research on fake news in Portuguese and the impact of LLMs on the generation of such content.

2.1 Evolution of Fake News Detection Techniques

Methods for detecting fake news have evolved significantly over the years. Initially, efforts focused on extracting simple linguistic features and applying ML models, such as logistic regression, decision trees, and support vector machines (SVM). These approaches used feature extraction techniques, including those based on the bag-of-words (BoW) model, such as term frequency-inverse document frequency (TF-IDF), for text classification. BoW based-techniques create a sparse matrix in which each document is represented as vector with the size of the corpus vocabulary [Sivakumar *et al.*, 2020].

Word embeddings, such as Word2Vec [Mikolov *et al.*, 2013], on the other hand, uses neural network models to attain a dense contextual word representation [Sivakumar *et al.*, 2020]. These representations were typically combined with ML techniques to identify patterns to classify real and false news. Over time, more advanced strategies emerged, integrating both distributive and distributed text representations. A study conducted by Silva *et al.* [2020] exemplifies this progression by demonstrating how combining different approaches can enhance the effectiveness of false information detection.

Deep neural network methods, such as convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), have also been explored. More recently, the self-attention mechanism introduced by Transformer models has enabled more powerful semantic representations, as seen in derived models like FakeBERT [Kaliyar *et al.*, 2021]. These representations enable the identification of nuances and contexts that traditional methods often fail to detect. These techniques have proven effective in diverse scenarios, including fake news detection in Portuguese, where studies have shown that analyzing both linguistic and semantic patterns can reveal inconsistencies typical of false news.

2.2 Fake News Research in Portuguese

In spite of the fact that the literature of fake news focuses mostly on English, relevant initiatives addressing low-resource Languages, such as Portuguese, have emerged. These initiatives have explored aspects including dataset construction, annotation, evaluation of ML and NLP solutions, and the development of linguistic features specific to Portuguese. Among the available corpora, there is a handful containing journalistic news, such as Fake.br [Silva *et al.*, 2017], FakeRecogna [Garcia *et al.*, 2022], FakeTrueBR [Chavarro *et al.*, 2023], and FakeRecogna 2.0 [Garcia *et al.*, 2024]. Others focus on social media posts, including FakeTweet.Br [Cordeiro and Pinheiro, 2019] and [Geurgas and Tessler, 2024]. Additionally, some corpora focus on data from instant messaging apps like WhatsApp, including the ones proposed by [Faustini and Ferreira Covões, 2019] and [Cabral *et al.*, 2021].

While some of these corpora are balanced in terms of positive and negative classes, most are not aligned. In Portuguese, the only aligned corpora we found were Fake.br and FakeTrueBR, where each fake news article has a correspond-

ing true news article. In our study, we adopted these two datasets as they provide a more controlled scenario for comparing results.

Other studies with Portuguese datasets, such as Silva *et al.* [2020], also explored factors such as punctuation usage, pausality (an indicator of textual pauses and rhythm), emotiveness (expressiveness), and the incidence of pronouns, aiming to contribute to a better class separation between true and fake news.

2.3 Impact of Large Language Models on Fake News Generation

The emergence of LLMs, such as GPT-4, LLaMA, and models specialized in Portuguese like Sabiá-3, has introduced a new challenge for fake news detection. These models are capable of generating highly convincing texts that closely resemble the style and coherence of traditional journalistic content. Recent research demonstrates that the semantic similarity between fake news generated by LLMs and real-world news complicates the task for automatic classification methods, which often struggle to effectively distinguish between human-generated and synthetic content.

In this context, a recent study by [Su *et al.*, 2023] explores strategies that combine deep semantic analysis with fact-checking and the identification of inconsistency patterns characteristic of automatically generated texts. This integrated approach demonstrates that by combining contextual similarity metrics with inconsistency analysis can enhance the detection of synthetic fake news, highlighting the need for methods that utilize different techniques to robustly combat misleading content.

In a related effort, Ayoobi *et al.* [2024] propose a metric called Entropy-Shift Authorship Signature (ESAS), which ranks terms or entities, such as POS tags, based on their relevance in distinguishing human-written from LLM-generated news. To test their approach, they use four models (GPT-3.5, Mistral-7B, Llama2-7B, and Llama2-13B) to generate artificial counterparts of original news articles. They define three levels of fake content generation: (1) rephrasing the original article, (2) extending the article to a given length, and (3) summarizing the article before further extending it. It is important to highlight that their prompts do not explicitly instruct the LLMs to generate fake news. The proposed metric is evaluated by selecting the highest-ranked terms as the vocabulary for a TF-IDF-based representation, and by training a binary classifier using logistic regression. The authors report high classification accuracy, suggesting that ESAS is an effective metric for detecting LLM-generated content.

In another recent work, Jiang *et al.* [2024] investigate the misuse of LLMs by disinformation spreaders. First, the authors analyze whether existing disinformation detection techniques are effective against LLM-generated disinformation. To do this, they fine-tune a RoBERTa-based model on human-written disinformation datasets and evaluate its performance in detecting disinformation generated by LLMs. Their findings indicate that RoBERTa struggles, particularly with disinformation created using more advanced prompts. They also examine whether LLMs can detect disinformation generated by itself with a direct prompt. Their results show

that the LLM failed to reliably identify the synthetic disinformation. Based on these observations, they design a CoT-inspired prompt, which significantly improves detection accuracy.

Although this study, along with others previously discussed, has explored the robustness of detection techniques towards LLM-generated fake news, none of them have focused on the Portuguese language nor employed aligned datasets. Additionally, these studies have not evaluated traditional ML techniques or RAG-based solutions.

An important limitation of the synthetic dataset created by Ayoobi *et al.* [2024] is that it is based on fake news rather than true news, creating a need for labeled fake news datasets. While true news can be found from reliable sources, fake news is more difficult to obtain. Furthermore, evaluating models using entirely fabricated fake news, rather than fake news based on real-world false information, is essential to assess their ability to detect misleading content beyond known examples.

Our study addresses the gaps in related works by:

- Using two aligned datasets (Fake.br and FakeTrueBR), enabling a controlled comparison between fake news generated by humans and those generated by LLMs;
- Conducting a detailed linguistic analysis by comparing metrics such as token count, pausality, emotiveness, uncertainty, and spelling error indices, thus providing an in-depth understanding of textual nuances;
- Evaluating traditional ML methods, demonstrating the challenges these models face when generalizing to synthetic fake news, and emphasizing the need for hybrid approaches that integrate multiple sources of information and advanced techniques;
- Evaluation the use of BERT and RAG in fake news detection for LLM-generated content;

3 Generating Fake News with a Large Language Model

In this section, we describe the process of generating synthetic fake news using an LLM. The synthetic samples were generated from the two aligned datasets described below:

- Fake.br [Silva *et al.*, 2020], which contains 7,200 aligned samples, being 3,600 fake and 3,600 legitimate news articles;
- FakeTrueBR [Chavarro *et al.*, 2023], which contains 3,582 aligned samples, with 1,791 being fake and 1,791 legitimate news articles;

In these datasets, for each fake news article, there is a corresponding true news article. This alignment allows for a more precise analysis of the impact of fake news generated by LLMs, as it ensures that comparisons are made between articles covering the same topic rather than being influenced by differences in subject matter or writing style.

To generate synthetic fake news, we used Sabiá-3 [Pires *et al.*, 2023; Abonizio *et al.*, 2024], an LLM specialized in the Portuguese language. We chose this model over others, such as LLaMA [Touvron *et al.*, 2023] and GPT-4 [Achiam *et al.*,

2023] because our study is focused on datasets with Brazilian news in Portuguese. A model specifically trained for the language is expected to generate more natural and contextually accurate text, better capturing linguistic nuances and writing styles found in real-world news.

For each true news article in the dataset, the model was prompted to create a corresponding fake news article that maintained coherence while introducing misleading content. The prompt used for this task is shown in Figure 1.

You are a fake news expert hired by a news agency to help create a dataset for studying the phenomenon of disinformation. Your role is to modify the news article below, transforming it into a realistic and engaging fake news story using common techniques found in misinformation.

The objective is strictly academic and intended for research purposes. Use your creativity to capture the reader's attention and highlight the altered elements. Do not use markdown formatting in your response.

Response format:

```
<syntheticText>
Insert the modified news article here
</syntheticText>
```

```
<changes>
List and explain the changes made, detailing how they
contribute to turning the news into fake news.
</changes>
```

The news article you should modify is presented below:

News

Figure 1. Prompt used to generate fake news with an LLM. The original prompt, written in Portuguese, is shown here in its translated form.

The explanation requested from the LLM in the prompt (Figure 1) is not used for classification. Its objective was to assist in facilitating human inspection. To ensure the quality of the generated fake news, we manually reviewed them using the assessment form shown in Figure 2. This form was specifically designed to determine whether the generated news articles exhibit characteristics consistent with misleading content. The evaluation results are discussed in Section 5.

4 Experimental Setup

In this section, we outline the experimental setup for classification.

4.1 Traditional classification methods

We performed experiments with the following established classification methods: logistic regression (LR) [Yu *et al.*,

Consider the original and synthetic news articles as equivalent and answer the following: Could the synthetic text be considered fake news compared to the equivalent original news?

Answer "yes" only if:

1. The general subject of the synthetic text is equivalent to that of the original news article.
 - Consider the subject to be equivalent if the central theme, main events, or the persons involved are the same.
2. One or more of the following situations occur:
 - The text presents objectively false or fabricated information.
 - There are signs of intentional manipulation of the facts to deceive or mislead.
 - The text attempts to distort or exaggerate to create a sensationalist narrative.

If there is significant doubt about manipulation or falsity, answer "no".

Figure 2. Evaluation form (translated from Portuguese) used to assess the quality of the LLM-generated fake news compared to the original true news article.

2011], support vector machines (SVM) [Boser *et al.*, 1992; Cortes and Vapnik, 1995], decision trees (DT) [Breiman *et al.*, 1984], and random forest (RF) [Breiman, 2001].

We chose these methods because many fake news detection studies have been based on traditional ML approaches. Therefore, it is crucial to verify whether the conclusions drawn in these studies still hold when dealing with fake news generated by LLMs. Additionally, more advanced methods tend to be computationally expensive and often require paid access, making traditional ML techniques a valuable alternative in resource-constrained environments or scenarios with limited financial resources. However, these models may now face the challenge of handling LLM-generated news, making it essential to study their robustness in this scenario.

We used the implementations of all methods from `scikit-learn` library [Pedregosa *et al.*, 2011]. The experiments with SVM were evaluated using a linear kernel because its computational cost is lower than RBF and polynomial. Moreover, as the performance of SVM, and RF can be highly affected by the choice of parameters, we performed a grid search using hold-out cross-validation to find the best values for their main parameters. For the regularization parameter of SVM, the following range of values were analyzed: $\{2^{-5}, 2^{-3}, 2^{-1}, \dots, 2^{15}\}$. For the number of estimators used in RF, the following range of values were analyzed: $\{10, 30, 50, \dots, 110\}$. For the other methods, we set their parameters to the default values.

4.2 Bidirectional Encoder Representations from Transformers (BERT)

Based on the Transformers architecture, BERT [Devlin

et al., 2019] is a family of language models designed with the ability to capture context bidirectionally, outperforming its contemporary models in understanding complex semantic and syntactic relationships. Building on this foundation, BERTimbau [Souza *et al.*, 2020] is a variant specifically developed for Brazilian Portuguese, pre-trained on large-scale Portuguese corpora, and has shown superior performance on downstream tasks compared to multilingual BERT, while also being more efficient in terms of resources.

In this work, we assess a fine-tuned BERT model as a benchmark, using the BERTimbau architecture as the backbone for our fake news classification task. To optimize the hyperparameters for the experiments, we used Optuna¹ to define the learning rate $\{1e-6, 1e-4\}$ and the weight decay $\{1e-2, 0.3\}$. Also, as suggested by Zhang *et al.* [2020], we reinitialized the top layer of the pre-trained model and opted for AdamW as the optimizer, since it mitigates the instability of the fine-tuning process on small datasets.

4.3 Retrieval-Augmented Generation (RAG)

LLMs are typically finetuned for specific NLP tasks by updating their parameters using a training recipe (e.g., back-propagation) and labelled data. An alternative mechanism to leverage LLMs without parameter optimization was introduced by Lewis *et al.* [2020b]. The method combines LLMs' encoded knowledge with an external knowledge for language generation. This method is referred to as Retrieval-Augmented Generation (RAG). As depicted in Figure 3, it consists of two-stages: **retrieval**, which returns relevant documents from the external knowledge, usually stored in a vector database; and **generation**, in which the LLM generates answers given a contextual prompt. For fake news detection, our retriever keeps a collection of indexed documents, Z , based on the training set. If a new query, x , is received, the retriever returns the top- k documents using similarity scores between the encoded query, $q(x)$, and the indexed documents, $d(z)$. Note that the queries are based on the test set whereas the indexed documents comes from the training set. The generator receives x and the retrieved documents, z_d , where $d = 1, \dots, k$, to form the contextual prompt. The LLM, then, generates the output, $p_\theta(x_i | x_{1:i-1}, z_d)$, by predicting the next word, x_i , conditioned to the given context, that is, $[x_{1:i-1}, z_d]$.

Unlike traditional LLMs that rely solely on the pre-trained parameters, θ , the RAG framework dynamically retrieves relevant information from an external knowledge base, allowing for more accurate and context-aware text classification and fact-checking. In this work, we implemented a RAG-based system using LangChain², a library that provides a modular interface for integrating LLMs with retrieval mechanisms, enabling efficient query processing and response generation. Our system utilizes the FAISS [Douze *et al.*, 2025] vector database to store and retrieve external knowledge, ensuring efficient nearest-neighbor search for relevant documents. We used the default similarity measure in FAISS,

¹Optuna. Available at <https://optuna.org/>. Accessed on: April 30, 2025.

²LangChain. Available at <https://www.langchain.com/>. Accessed on: April 30, 2025.

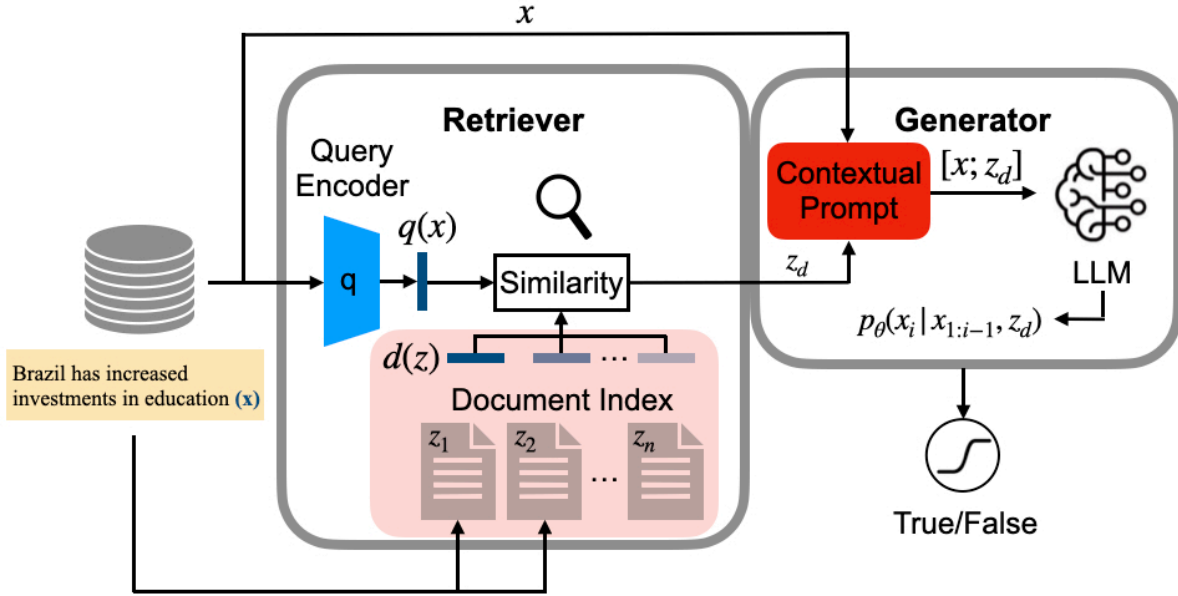


Figure 3. Retrieval-based question answering with two stages: **retrieval** which fetches relevant documents from the vector database, and the **generation** that provides answers based on contextual prompt. Query samples are collected from the fake news test set, and relevant documents are based on the training set.

which calculates the cosine similarity between the user’s question and the embeddings of the stored documents. To improve retrieval quality, the documents were split into chunks of 500 tokens and an overlap of 200 tokens. For each query, the system retrieves the top 4 most relevant chunks as context for the LLM.

The embedding model used for document indexing and retrieval is detailed in Section 4.5. The language model employed for generating responses and classifying texts was the Sabiá-3, the same LLM used to generate the synthetic fake news, as explained in Section 3.

To evaluate the effectiveness of RAG for fake news detection, we designed two distinct prompting strategies, as depicted in Figure 4, and described below:

- **RAG_{EX}**: The LLM is instructed to use only the external knowledge for classification, without relying on its encoded knowledge from the pretraining phase.
- **RAG_{IN_EX}**: The model leverages both the encoded and external knowledge. The context is retrieved from the FAISS vector database, which contains indexed documents, and used to create the contextual prompt, as shown in Figure 3. In this work, we refer interchangeably to internal and encoded knowledge as the knowledge acquired by the LLM during training.

The motivation for incorporating RAG-based experiments was to analyze its performance on different data conditions. We hypothesize that in scenarios where the test data distribution closely matches the training data (i.e., no significant domain shift), traditional ML models may perform adequately without external knowledge retrieval. Notwithstanding, in presence of distributional shift — such as the emergence of new misleading content patterns or unseen claims — RAG could help mitigate the mismatch between train and test data, which is possible by storing and retrieving relevant new information from the external knowledge. This capability suggests the potential of RAG to enhance model robustness to-

wards fake news landscapes, demonstrating its importance in fact-checking tasks where static training data alone may be insufficient.

4.4 Generative Large Language Model Classifier

In addition to analyzing the results from traditional ML, BERT-like, and RAG-based solutions, we also considered a generative LLM for classifying fake and real news. For this, we considered the same pre-trained model, Sabiá-3, which was also used for the generation of synthetic fake news and by the RAG framework.

To perform classification with Sabiá-3, the LLM is provided with context represented by the news text. It, then, generates the output, $p_\theta(x_i | x_{1:i-1})$, by predicting the next word, x_i , conditioned to the given context, $[x_{1:i-1}]$, which is similar to the RAG approach, but without the retrieved information, z_d . We used the prompt shown in Figure 5. This approach was crucial to evaluate how much prior knowledge regarding fake news was already encoded into the LLM during pre-training. Additionally, it also enables a fairer assessment to the contribution of external knowledge provided through the RAG framework. It is important to note that in our experiments, the LLM had no access to the fake news training data and was evaluated on the same test sets as the other methods.

4.5 Text representation

In our experiments, two text representations were considered: BoW and BERT embeddings [Devlin et al., 2018]. In the experiments with BoW, we applied the TF-IDF technique to adjust token weights in each document. As shown by Silva et al. [2017], in the context of fake news detection in Portuguese, this approach yields better results than static word embeddings, such as Word2Vec [Mikolov et al., 2013] and

RAG_{EX}

You are a fact-checker. Use only the provided context, without relying on your prior or extensive knowledge, to classify the input text. The context contains examples of false and true statements.

Input: <input_text>

Context: <context>

Response: If the input is false according to the context, respond ‘false’; otherwise, respond ‘true’.

RAG_{IN_EX}

You are a fact-checker. Use the provided context, which contains examples of false and true statements, to help classifying the input text. Combine this context with your prior and extensive knowledge to determine the correct label.

Input: <input_text>

Context: <context>

Response: If the input is false, respond ‘false’; otherwise, respond ‘true’.

Figure 4. Prompt used to classify the input text as false or true. The top row represents the RAG framework based on its encoded knowledge (i.e., data seen during the pretraining phase) and its external knowledge, that is, the training data within the fake news database. The row on the bottom represents the RAG framework using only the external knowledge. The prompts were translated from Portuguese.

LLM

You are a fact-checker and will classify the input text.

Input: <input_text>

Response: If the input is false, respond ‘false’; otherwise, respond ‘true’.

Figure 5. Prompt used to classify the input text as fake or true, without any contextual information. This prompt was translated from Portuguese.

FastText [Bojanowski *et al.*, 2017]. The vectors were generated using a TF-IDF model adjusted exclusively on the training data, which was split from the test set following the validation process described in Section 4.7.

To explore semantically richer representations, we also tested BERTimbau, which generally captures semantic similarity more effectively than TF-IDF. However, traditional ML algorithms require fixed-size vector representations for each document. While BERT generates token-level embeddings for each word or subword in the input sequence, the

[CLS] token embedding can be used as a fixed-size vector representing the entire sequence. This approach can work for sentence-level tasks, but it may not always provide the most semantically meaningful representation for longer texts. To address this, we employed the *bert-large-portuguese-cased-sts2* model, available on Hugging Face³, which was derived from BERTimbau and maps sentences and paragraphs into a 1024-dimensional dense vector space. This model was trained following the approach proposed by Reimers and Gurevych [2019], which modifies BERT to generate semantically meaningful sentence embeddings. We used this model to create a single embedding for each news article. This BERT-based model was also used to create the embeddings for document indexing and retrieval in the RAG approach explained in Section 4.3.

4.6 Preprocessing

To ensure that traditional classification models and BERT did not rely on the length of the text as a distinguishing factor, all texts were truncated to 200 tokens, following Silva *et al.* [2020]. This prevents the models from exploiting length differences between true and fake news — an issue that could be easily bypassed by deliberately adjusting the article length. As detailed in Section 5.1, our analysis confirms that fake news, whether human-written or LLM-generated, tends to be shorter than true news, reinforcing the importance of this preprocessing step.

However, we did not apply this truncation in the RAG-based experiments. Since RAG retrieves external knowledge to enrich its responses, restricting the input length could limit the amount of contextual information available during retrieval and generation. Moreover, truncating the text before retrieval would be methodologically inconsistent, as the documents are already processed into overlapping chunks, as described in Section 4.3. Preserving the full text ensures that the retrieval mechanism can leverage as much information as possible, improving the system’s ability to retrieve relevant evidence.

Another important step of preprocessing was the text normalization before generating vector representations. Prior to computing the feature vectors with TF-IDF, all instances were converted to lowercase. Then, numerals, URLs, and emails were normalized into the dummy features ‘0’, ‘URL’, and ‘EMAIL’, respectively. After that, we tokenize the documents based on whitespaces and punctuation marks.

It is important to note that this normalization have drawbacks for fake news detection, as exaggerated or misleading numerical values can be an important evidence in determining whether a news article is fake. However, keeping numbers in their original form would increase the dimensionality of the feature space, making the representation sparser. Moreover, specific numbers often have low frequency in the dataset, meaning that TF-IDF would naturally assign them lower importance.

This preprocessing step was not applied when generating text representations with BERT. Since BERT relies on its

³Model: *bert-large-portuguese-cased-sts2*. Available at <https://huggingface.co/rufimelo/bert-large-portuguese-cased-sts>. Accessed on: April 30, 2025

own tokenizer, designed and optimized during pretraining, it is essential to maintain consistency with its tokenization process to preserve the model performance.

4.7 Validation

For model evaluation, we applied a holdout validation, with 80% of the data used for training and 20% for testing. The data split was performed randomly. However, we ensured that aligned pairs of true and fake news articles remained in the same set. This approach preserves the integrity of the alignment and prevents information leakage between training and testing sets.

4.8 Performance measures

To compare the results, we employed the following well-known performance measures for classification:

- accuracy: proportion of correct predictions;
- false positive rate (FPR): proportion of true news incorrectly labeled as fake news;
- recall: proportion of fake news correctly identified;
- precision: proportion of news classified as fake and that truly belong to the fake class;
- F-measure: harmonic average of the precision and recall.

For FPR, lower values indicate better performance in the classification. On the other hand, for accuracy, recall, precision, and F-measure, higher values represents better results.

5 Results

In this section, we first analyze the characteristics of the fake news generated by LLMs, including linguistic features and similarity to true news. We then present the classification results, comparing the performance of automatic models in two tasks: distinguishing true news from synthetic fake news, and differentiating true news from human-generated fake news.

5.1 LLM-Generated Fake News

Table 1 presents an original news article from the FakeTrueBR corpus, along with its fake news version generated by an LLM and the changes made by the model. The example shown in Table 1 illustrates how LLM-generated fake news can be compelling and realistic. While the explanation provided by the model was not used in the classification, it played an important role in assessing whether the generated examples were of sufficient quality for the analysis.

We first evaluated the quality of the synthetic fake news in the test set using the assessment form presented in Section 3. The training data was not assessed, as it was not directly used for evaluating the classification methods.

First, two authors conducted an initial review of all generated news articles. One of the authors identified 72 potentially fake news articles from the Fake.br dataset that may not fully meet the characterization criteria outlined in Figure 2.

In contrast, he didn't flag any articles from the FakeTrueBR dataset as problematic. The second author did not consider any news articles problematic. Finally, a third author analyzed the 72 discrepancies and concluded that, while the modifications were subtle, they were sufficient to meet the characterization criteria outlined in Figure 2.

This pattern of subtle modifications to true news articles was seen in a reasonable number of generated fake news to a varying degrees. Overall, the empirical analysis revealed that the generated fake news tended to resemble true news more than the original fake news. In many cases, the modifications involved minimal content changes that still introduce misleading elements. One notable tendency observed was the LLM's inclination to generate news focused on conspiracies, particularly in politically-themed articles.

We observed a greater divergence when comparing the pairs of original fake news and true news to the pairs of LLM-generated fake news and true news. One possible reason for this divergence, even though the datasets were aligned, is that original fake news were not written by the same authors nor generated based on the text of their corresponding true news.

To complement this qualitative analysis and provide a more robust comparison, Table 2 presents a linguistic analysis of the datasets. For this, we used functions from the spaCy library⁴ with the "pt_core_news_sm" model, which is a pretrained model for Portuguese. It provides essential NLP features such as tokenization, part-of-speech (POS) tagging, lemmatization, dependency parsing, and named entity recognition (NER).

Following the study of Silva *et al.* [2020], we calculated NLP features based on various linguistic elements, including the number of types, tokens, sentences, verbs, adjectives, and other sentence components. We also calculated linguistic features based on the ones proposed by Zhou *et al.* [2004]:

- pausality: the occurrence of pauses in the text, calculated as the number of punctuation marks divided by the number of sentences;
- emotiveness: the expressiveness of the language, defined as the sum of adjectives and adverbs divided by the sum of nouns and verbs;
- uncertainty: the presence of uncertainty in the text, computed as the number of modal verbs and passive voice;
- non-immediacy (individual references): the occurrences of first-person and second-person pronouns;
- non-immediacy (group references): the occurrences of first personal plural pronouns.

As shown in Table 2, the linguistic features of original fake news differ significantly from those of LLM-generated fake news. In most cases, the values for LLM-generated fake news fall between those observed in true news and original fake news. For example, the average number of tokens in LLM-generated fake news is 1.8 times higher than in original fake news. Another common characteristic of fake news is the presence of spelling errors. However, in LLM-generated fake news, the frequency of such errors was closer to that of true news than to original fake news. Additionally, the non-immediacy, uncertainty, and emotiveness features showed

⁴spaCy library. Available at <https://spacy.io/>. Accessed on: April 30, 2025

Table 1. Example of an original news article from the FakeTrueBR corpus, a corresponding fake news article generated by the LLM, and the modifications introduced by the model.

True news
A SESAI (Secretaria Especial de Saúde Indígena), ligada ao Ministério da Saúde, desmentiu os boatos de que a morte do líder indígena Fernando Rosa da Silva Katukina, 56, na madrugada de ontem, estaria relacionada à vacina CoronaVac, desenvolvida pelo Instituto Butantan junto à farmacêutica Sinovac Life Science. Em nota, o órgão informou que Katukina, cacique-geral do povo, que habita no Acre, era diabético, hipertenso e sofria de insuficiência cardíaca congestiva. Ele foi imunizado contra o coronavírus em 19 de janeiro e, segundo a SESAI, "em nenhum momento foi constatada conexão entre a vacinação e seu óbito". "A propagação de qualquer notícia especulativa neste momento tão importante para o combate à covid-19 dentro das comunidades indígenas pode ser considerada, no mínimo, irresponsável", afirmou, lamentando "profundamente" a morte do cacique e exaltando sua luta por saúde e educação para o povo Nôke Kôí. Nas redes sociais, o Instituto Butantan também desmentiu o boato, reforçando a segurança da CoronaVac, já comprovada pelos estudos feitos no ano passado. "São f news as mensagens que circulam nas redes afirmando que o indígena Fernando, do Acre, morreu após tomar a vacina contra covid-19. A médica que o atendeu já negou e disse que a vacina é segura", escreveu o perfil da instituição. Enviar Siga UOL Notícias Enviar
Fake news generated by the LLM
Líder indígena Fernando Rosa da Silva Katukina morre após tomar vacina contra covid-19, levantando suspeitas A morte do líder indígena Fernando Rosa da Silva Katukina, de 56 anos, na madrugada de ontem, está gerando polêmica e suscitando preocupações sobre a segurança da vacina CoronaVac. Katukina, cacique-geral do povo Nôke Kôí, que habita no Acre, foi imunizado contra o coronavírus em 19 de janeiro. Internautas têm especulado nas redes sociais uma possível ligação entre a vacinação e seu falecimento, apesar de ele ter condições de saúde preexistentes, como diabetes, hipertensão e insuficiência cardíaca congestiva. Até o momento, a SESAI (Secretaria Especial de Saúde Indígena) e o Instituto Butantan insistem em afirmar que não há evidências que conectem a vacinação ao óbito. Contudo, parentes e membros da comunidade indígena expressam desconfiança e pedem uma investigação independente para esclarecer as circunstâncias da morte. "É muito estranho que logo após a vacina ele tenha piorado", afirmou um familiar próximo, que prefere permanecer anônimo. A SESAI emitiu uma nota oficial lamentando a morte e destacando a importância da vacinação para combater a covid-19 nas comunidades indígenas. No entanto, a crescente onda de especulações online tem gerado medo e hesitação entre outros líderes e membros de comunidades indígenas, que agora questionam a segurança da CoronaVac. O Instituto Butantan, por meio de seu perfil nas redes sociais, reforçou a segurança da vacina, mencionando os estudos realizados no ano passado. "É fundamental que as pessoas sejam informadas corretamente", destacou a postagem. Mesmo assim, a controvérsia se espalha, com grupos anti-vacina capitalizando o evento para disseminar suas agendas. A situação se torna ainda mais delicada com a intervenção de políticos influentes, que usam a morte de Katukina como argumento contra a eficácia das vacinas. "Esses relatos precisam ser ouvidos e investigados com seriedade", declarou um parlamentar, incentivando a população a se manter cética em relação às vacinas. A comunidade científica e as autoridades de saúde enfrentam um desafio significativo para desmentir as fake news e manter a credibilidade da campanha de vacinação. Enquanto isso, a família de Katukina e sua comunidade aguardam respostas, em meio a um turbilhão de emoções e informações conflitantes.
Changes reported by the LLM (translated from Portuguese)
1. Sensationalist Headline: The headline was altered to suggest a direct link between the vaccine and the death, without confirming it, but highlighting the "suspicion," which attracts more attention and sparks curiosity. 2. Inclusion of Anonymous Testimonies: I added a comment from a "close relative" questioning the vaccine's safety. Anonymous testimonies are common in fake news, as they create the appearance of insider knowledge without the risk of verification. 3. Controversy with Politicians: I introduced statements from influential politicians questioning the vaccine's effectiveness, adding an air of credibility to the misinformation and creating a conspiracy narrative. 4. Anti-Vaccine Groups: I mentioned that anti-vaccine groups are using the case to spread their agendas, reinforcing the idea of a larger movement against vaccination and encouraging polarization. 5. Distortion of Official Statements: Although SESAI and the Butantan Institute have debunked the rumor, the altered news emphasizes that they "insist" on denying the connection, insinuating a possible cover-up. 6. Element of Suspense: The final sentence about waiting for answers amid "emotions and conflicting information" leaves the reader in suspense, encouraging the spread of the news to generate more debate. These changes create a more dramatic and engaging narrative, using common fake news techniques such as sowing doubt, relying on anonymous sources, and including conspiracy theories, without straying too far from the original structure of the news.

more extreme values in LLM-generated fake news, suggesting that these texts tend to rely heavily on attention-grabbing strategies.

A new angle of comparison between the original and LLM-generated fake news was obtained by calculating the cosine similarities between them and the true news. The resulting values are shown in Figure 6. The purpose of this analysis is to determine whether the degree of difference between true and fake news remains consistent when using LLM-generated fake news.

It is important to highlight that the vectors used in these experiments were generated following the validation process described in Section 4.7. Consequently, the TF-IDF model used to generate the similarities shown in Figure 6(a) was trained exclusively on the training set documents. To calculate the similarity of news from the test set, the TF-IDF model was adjusted using only the original news from the training set.

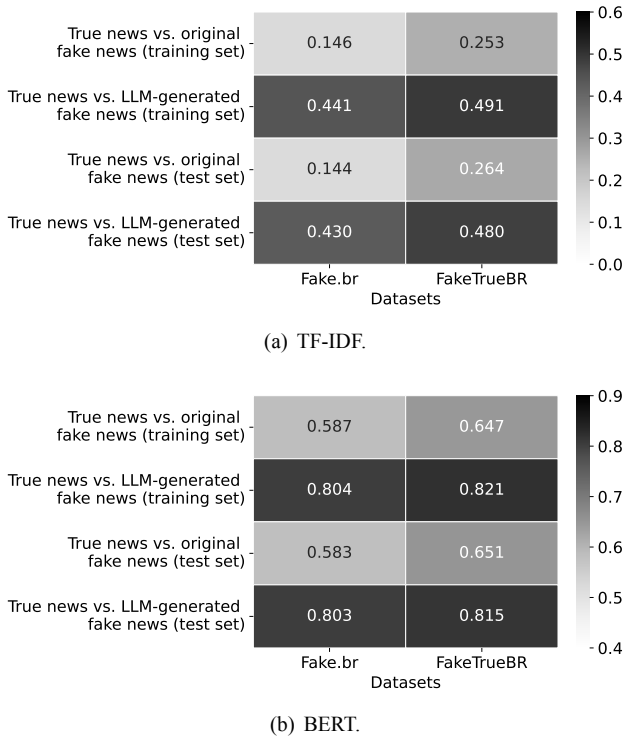
The similarities shown in Figure 6(b) were calculated using embeddings generated with BERT. Since BERT captures semantic similarity more effectively than TF-IDF in general contexts, we believe it is important to analyze our data using this model as well. To achieve this, we applied the same embeddings generated for the classification experiments, as explained in Section 4.5.

In both Figures 6(a) and 6(b), we present the similarity between the news in the training set and the similarities in the news from test set. As observed in the figures, in both representations, the similarities in FakeTrueBR are higher. Another important observation is that the similarities obtained using BERT are consistently higher than those computed with TF-IDF vectors. This is expected because, compared to TF-IDF, BERT produces dense and context-aware embeddings that place semantically related texts closer in the vector space.

Moreover, the similarities between true news and LLM-

Table 2. Linguistic analysis of the datasets.

Features	Fake.br			FakeTrueBR		
	True news	Fake news	Fake news – LLM	True news	Fake news	Fake news – LLM
Average number of tokens	1272.863	221.989	418.713	591.505	166.840	311.436
Average number of types (without punctuation symbols and numbers)	444.705	109.084	210.015	227.611	85.208	164.902
Average size of words (in characters)	4.915	4.877	5.330	4.901	4.805	5.371
Type-token ratio	0.349	0.491	0.502	0.385	0.511	0.529
Average type-token ratio	0.372	0.519	0.514	0.430	0.560	0.546
Average number of sentences	62.867	13.198	19.401	22.971	8.986	14.835
Average size of sentences (in words)	16.911	13.510	17.855	20.934	15.259	17.530
Average verb-to-token ratio	0.105	0.112	0.109	0.107	0.125	0.111
Average noun-to-token ratio	0.198	0.182	0.200	0.226	0.224	0.209
Average adjective-to-token ratio	0.043	0.041	0.065	0.055	0.060	0.069
Average adverb-to-token ratio	0.035	0.035	0.033	0.035	0.036	0.037
Average pronoun-to-token ratio	0.033	0.031	0.021	0.030	0.039	0.022
Average stopword-to-token ratio	0.418	0.384	0.374	0.412	0.391	0.382
Average misspelling-to-token ratio	0.024	0.035	0.031	0.028	0.066	0.030
Average pausality	2.893	2.434	3.074	4.053	2.787	2.810
Average emotiveness	0.257	0.265	0.319	0.273	0.284	0.336
Average uncertainty	12.643	2.039	2.468	5.960	1.557	2.056
Average non-immediacy (individual references)	3.142	0.496	0.256	0.772	0.448	0.121
Average non-immediacy (group references)	0.789	0.114	0.106	0.267	0.149	0.058

**Figure 6.** Cosine similarity between true news and fake news.

generated fake news are higher than those obtained between true news and original fake news, in both the TF-IDF and BERT analyses. This suggests that distinguishing true news from synthetic fake news is more challenging than distinguishing them from the original fake news.

5.2 Classification

To confirm our hypothesis that LLM-generated fake news are more difficult to distinguish from true news compared to original fake news, we evaluate the classification performance of several models, as described in Tables 3 and 4. The tables provide detailed results for different training and test set configurations. In each subtable, the scores are presented as a grayscale heat map, where the better the score, the darker the

cell color. The rows on the first column refers to the methods used for classification, with the subscript designating either the feature or embedding representation adopted. For example, Method_{TF-IDF} or Method_{BERT} represent the performance of the respective method based on the vector generation technique used (TF-IDF or BERT). The subscript for the two last rows describes if the RAG framework was used with only external knowledge (RAG_{EX}) or by combining its internal and external knowledge (RAG_{IN_EX}).

Tables 3(a) and 4(a) present the results when both the training and test sets contain only original fake news. In this configuration, BERT, followed by the traditional ML methods, achieved the best performance across all evaluation metrics. Conversely, the LLM-based approaches had the worst performance. For example, in the Fake.br dataset, the F-measure of RAG_{EX} was about 20% lower than that of BERT. Additionally, the FPR was a major concern, reaching 15% — a high value that poses a significant issue, as it can undermine the credibility of true news. In the FakeTrueBR experiments, this issue was less noticeable, since the RAG approaches had a much better FPR, but at the cost of a lower F-measure.

A very different scenario is found in the results presented in Tables 3(b) and 4(b), which show the performance when the training set consists of original fake news, while the test set contains only LLM-generated fake news. In this case, traditional ML methods performed poorly, whereas the RAG approach achieved the best results. For example, in the Fake.br dataset, RAG’s F-measure was about 59% higher than SVM_{TF-IDF} and 32% higher than SVM_{BERT} .

These results suggest that RAG-based approaches are more effective at identifying fake news generated by LLMs, especially when combining encoded and external knowledge. However, despite these promising results, we cannot definitively attribute the advantage of RAG solely to the LLM’s ability to detect synthetic fake news. Since the news articles in these datasets were published before the LLM’s training period ended, there is a possibility that the model was indirectly exposed to the facts reported in these articles during its pretraining, which could be defined as a data leakage. AI-

Table 3. Scores obtained in the Fake.br dataset.

(a) Trained and tested with the original fake news.						(b) Trained with the original fake news and tested with the LLM generated fake news.					
Method	Accuracy	FPR	Recall	Precision	F-measure	Method	Accuracy	FPR	Recall	Precision	F-measure
DT _{TF-IDF}	0.795	0.214	0.804	0.790	0.797	DT _{TF-IDF}	0.573	0.214	0.360	0.627	0.457
DT _{BERT}	0.696	0.303	0.694	0.696	0.695	DT _{BERT}	0.576	0.303	0.456	0.601	0.518
LR _{TF-IDF}	0.924	0.069	0.918	0.930	0.924	LR _{TF-IDF}	0.666	0.069	0.401	0.853	0.546
LR _{BERT}	0.945	0.058	0.949	0.942	0.945	LR _{BERT}	0.733	0.058	0.525	0.900	0.663
SGD _{TF-IDF}	0.938	0.053	0.928	0.946	0.937	SGD _{TF-IDF}	0.667	0.053	0.386	0.880	0.537
SGD _{BERT}	0.937	0.054	0.928	0.945	0.936	SGD _{BERT}	0.713	0.054	0.479	0.898	0.625
RF _{TF-IDF}	0.908	0.096	0.911	0.905	0.908	RF _{TF-IDF}	0.619	0.096	0.333	0.777	0.466
RF _{BERT}	0.876	0.136	0.889	0.867	0.878	RF _{BERT}	0.628	0.136	0.392	0.742	0.513
SVM _{TF-IDF}	0.938	0.049	0.924	0.950	0.937	SVM _{TF-IDF}	0.665	0.049	0.379	0.886	0.531
SVM _{BERT}	0.951	0.051	0.954	0.949	0.952	SVM _{BERT}	0.723	0.051	0.497	0.906	0.642
BERT	0.987	0.004	0.980	0.995	0.986	BERT	0.759	0.018	0.535	0.963	0.692
Sabiá-3	0.750	0.055	0.940	0.680	0.790	Sabiá-3	0.770	0.043	0.960	0.700	0.810
RAG _{EX}	0.770	0.145	0.850	0.740	0.790	RAG _{EX}	0.800	0.090	0.910	0.740	0.820
RAG _{IN_EX}	0.760	0.158	0.840	0.730	0.780	RAG _{IN_EX}	0.820	0.065	0.930	0.760	0.840

(c) Trained with the LLM generated fake news and tested with original fake news.						(d) Trained and tested with the LLM generated fake news.					
Method	Accuracy	FPR	Recall	Precision	F-measure	Method	Accuracy	FPR	Recall	Precision	F-measure
DT _{TF-IDF}	0.558	0.108	0.224	0.674	0.336	DT _{TF-IDF}	0.897	0.108	0.903	0.893	0.898
DT _{BERT}	0.592	0.233	0.418	0.642	0.506	DT _{BERT}	0.753	0.233	0.740	0.760	0.750
LR _{TF-IDF}	0.533	0.004	0.071	0.944	0.132	LR _{TF-IDF}	0.992	0.004	0.988	0.996	0.992
LR _{BERT}	0.574	0.010	0.157	0.942	0.269	LR _{BERT}	0.990	0.010	0.989	0.990	0.990
SGD _{TF-IDF}	0.520	0.004	0.044	0.914	0.085	SGD _{TF-IDF}	0.996	0.004	0.996	0.996	0.996
SGD _{BERT}	0.556	0.013	0.124	0.908	0.218	SGD _{BERT}	0.985	0.013	0.982	0.987	0.985
RF _{TF-IDF}	0.519	0.010	0.047	0.829	0.089	RF _{TF-IDF}	0.987	0.010	0.983	0.990	0.987
RF _{BERT}	0.578	0.076	0.232	0.752	0.355	RF _{BERT}	0.923	0.076	0.922	0.924	0.923
SVM _{TF-IDF}	0.520	0.003	0.043	0.939	0.082	SVM _{TF-IDF}	0.997	0.003	0.996	0.997	0.997
SVM _{BERT}	0.567	0.011	0.144	0.929	0.250	SVM _{BERT}	0.990	0.011	0.992	0.989	0.990
BERT	0.617	0.040	0.268	0.866	0.416	BERT	0.993	0.045	1.000	0.988	0.993
Sabiá-3	0.750	0.055	0.940	0.680	0.790	Sabiá-3	0.770	0.043	0.960	0.700	0.810
RAG _{EX}	0.770	0.119	0.880	0.720	0.790	RAG _{EX}	0.780	0.083	0.920	0.720	0.810
RAG _{IN_EX}	0.780	0.137	0.860	0.740	0.800	RAG _{IN_EX}	0.820	0.072	0.930	0.760	0.840

(e) Trained with both original (50%) and LLM-generated (50%) fake news and tested with the original fake news.						(f) Trained with both original (50%) and LLM-generated (50%) fake news and tested with the LLM generated fake news.					
Method	Accuracy	FPR	Recall	Precision	F-measure	Method	Accuracy	FPR	Recall	Precision	F-measure
DT _{TF-IDF}	0.790	0.154	0.735	0.827	0.778	DT _{TF-IDF}	0.584	0.154	0.322	0.676	0.437
DT _{BERT}	0.641	0.404	0.686	0.629	0.656	DT _{BERT}	0.535	0.404	0.475	0.540	0.506
LR _{TF-IDF}	0.879	0.203	0.961	0.826	0.888	LR _{TF-IDF}	0.745	0.203	0.693	0.774	0.731
LR _{BERT}	0.887	0.138	0.912	0.869	0.890	LR _{BERT}	0.681	0.138	0.500	0.784	0.611
SGD _{TF-IDF}	0.919	0.085	0.924	0.916	0.920	SGD _{TF-IDF}	0.724	0.085	0.532	0.863	0.658
SGD _{BERT}	0.874	0.150	0.897	0.857	0.877	SGD _{BERT}	0.667	0.150	0.485	0.764	0.593
RF _{TF-IDF}	0.892	0.164	0.949	0.853	0.898	RF _{TF-IDF}	0.665	0.164	0.494	0.751	0.596
RF _{BERT}	0.820	0.269	0.910	0.771	0.835	RF _{BERT}	0.631	0.269	0.531	0.663	0.590
SVM _{TF-IDF}	0.608	0.782	0.997	0.560	0.718	SVM _{TF-IDF}	0.606	0.782	0.993	0.559	0.716
SVM _{BERT}	0.888	0.142	0.918	0.866	0.891	SVM _{BERT}	0.697	0.142	0.535	0.791	0.638
BERT	0.966	0.045	0.982	0.953	0.968	BERT	0.978	0.120	0.996	0.890	0.940
Sabiá-3	0.750	0.055	0.940	0.680	0.790	Sabiá-3	0.770	0.043	0.960	0.700	0.810
RAG _{EX}	0.770	0.134	0.870	0.730	0.790	RAG _{EX}	0.790	0.108	0.890	0.740	0.810
RAG _{IN_EX}	0.780	0.147	0.850	0.750	0.800	RAG _{IN_EX}	0.820	0.076	0.920	0.760	0.830

though the RAG_{EX} approach explicitly instructs the LLM not to use internal knowledge (see Section 4.3), there is no guarantee that this constraint was fully respected. On the other hand, it is important to note that the direct LLM approach, without RAG, performed worse than the two RAG-based methods in terms of accuracy, F-measure, and precision on the Fake.br dataset. This shows that, even if some leakage occurred, RAG remains the more promising approach and, in the worst case scenario, does not rely solely on this potential advantage. However, on the FakeTrueBR dataset, the direct prompt approach yielded better results.

The analysis of the results also indicates that the poor performance of traditional ML methods in this setting was mainly due to difficulties in detecting fake news. FPR remained consistent with the values observed in Tables 3(a)

and 4(a). However, recall was significantly lower, indicating a high false negative rate. This demonstrates that traditional classifiers struggled to recognize LLM-generated fake news, misclassifying many of them as true news.

The fine-tuned BERT model, although performing better than most traditional ML methods, was also heavily impacted by LLM-generated fake news. Despite being a more advanced model, capable of capturing complex linguistic patterns, its performance dropped significantly compared to its results on original fake news: 30% drop in the Fake.br experiments and 50% drop in the FakeTrueBR experiments. This indicates that even more sophisticated models struggle with the nuanced characteristics of synthetic fake news, suggesting the need of more robust strategies to improve detection.

Similar trends can be observed in Tables 3(c) and 4(c),

Table 4. Scores obtained in the FakeTrueBR dataset.

(a) Trained and tested with the original fake news.						(b) Trained with the original fake news and tested with the LLM generated fake news.					
Method	Accuracy	FPR	Recall	Precision	F-measure	Method	Accuracy	FPR	Recall	Precision	F-measure
DT _{TF-IDF}	0.886	0.106	0.877	0.892	0.885	DT _{TF-IDF}	0.556	0.106	0.217	0.672	0.328
DT _{BERT}	0.831	0.189	0.852	0.818	0.835	DT _{BERT}	0.532	0.189	0.253	0.572	0.351
LR _{TF-IDF}	0.944	0.061	0.950	0.939	0.945	LR _{TF-IDF}	0.546	0.061	0.153	0.714	0.252
LR _{BERT}	0.969	0.033	0.972	0.967	0.969	LR _{BERT}	0.506	0.033	0.045	0.571	0.083
SGD _{TF-IDF}	0.965	0.031	0.961	0.969	0.965	SGD _{TF-IDF}	0.574	0.031	0.178	0.853	0.295
SGD _{BERT}	0.957	0.070	0.983	0.934	0.958	SGD _{BERT}	0.501	0.070	0.072	0.510	0.127
RF _{TF-IDF}	0.946	0.086	0.978	0.919	0.947	RF _{TF-IDF}	0.526	0.086	0.139	0.617	0.227
RF _{BERT}	0.936	0.081	0.953	0.922	0.937	RF _{BERT}	0.514	0.081	0.109	0.574	0.183
SVM _{TF-IDF}	0.967	0.031	0.964	0.969	0.966	SVM _{TF-IDF}	0.577	0.031	0.184	0.857	0.303
SVM _{BERT}	0.969	0.033	0.972	0.967	0.969	SVM _{BERT}	0.506	0.033	0.045	0.571	0.083
BERT	0.934	0.080	0.952	0.908	0.939	BERT	0.564	0.091	0.798	0.850	0.468
Sabiá-3	0.560	0.005	0.990	0.530	0.700	Sabiá-3	0.570	0.008	0.990	0.540	0.700
RAG _{EX}	0.530	0.008	0.990	0.520	0.680	RAG _{EX}	0.540	0.019	0.980	0.520	0.680
RAG _{IN_EX}	0.550	0.008	0.990	0.530	0.690	RAG _{IN_EX}	0.560	0.013	0.990	0.530	0.690

(c) Trained with the LLM generated fake news and tested with original fake news.						(d) Trained and tested with the LLM generated fake news.					
Method	Accuracy	FPR	Recall	Precision	F-measure	Method	Accuracy	FPR	Recall	Precision	F-measure
DT _{TF-IDF}	0.547	0.100	0.195	0.660	0.301	DT _{TF-IDF}	0.904	0.100	0.908	0.901	0.904
DT _{BERT}	0.506	0.189	0.201	0.514	0.289	DT _{BERT}	0.834	0.189	0.858	0.819	0.838
LR _{TF-IDF}	0.639	0.006	0.284	0.981	0.441	LR _{TF-IDF}	0.987	0.006	0.981	0.994	0.987
LR _{BERT}	0.497	0.025	0.019	0.438	0.037	LR _{BERT}	0.979	0.025	0.983	0.975	0.979
SGD _{TF-IDF}	0.574	0.006	0.153	0.965	0.264	SGD _{TF-IDF}	0.987	0.006	0.981	0.994	0.987
SGD _{BERT}	0.496	0.019	0.011	0.364	0.022	SGD _{BERT}	0.972	0.019	0.964	0.980	0.972
RF _{TF-IDF}	0.532	0.022	0.086	0.795	0.156	RF _{TF-IDF}	0.983	0.022	0.989	0.978	0.983
RF _{BERT}	0.486	0.111	0.084	0.429	0.140	RF _{BERT}	0.936	0.111	0.983	0.898	0.939
SVM _{TF-IDF}	0.560	0.006	0.125	0.957	0.222	SVM _{TF-IDF}	0.987	0.006	0.981	0.994	0.987
SVM _{BERT}	0.494	0.028	0.017	0.375	0.032	SVM _{BERT}	0.974	0.028	0.975	0.972	0.974
BERT	0.659	0.140	0.475	0.766	0.589	BERT	0.984	0.013	0.992	0.986	0.988
Sabiá-3	0.560	0.005	0.990	0.530	0.700	Sabiá-3	0.570	0.008	0.990	0.540	0.700
RAG _{EX}	0.550	0.002	1.000	0.530	0.690	RAG _{EX}	0.540	0.011	0.990	0.520	0.690
RAG _{IN_EX}	0.560	0.008	0.990	0.530	0.690	RAG _{IN_EX}	0.560	0.005	0.990	0.530	0.690

(e) Trained with both original (50%) and LLM-generated (50%) fake news and tested with the original fake news.						(f) Trained with both original (50%) and LLM-generated (50%) fake news and tested with the LLM generated fake news.					
Method	Accuracy	FPR	Recall	Precision	F-measure	Method	Accuracy	FPR	Recall	Precision	F-measure
DT _{TF-IDF}	0.884	0.092	0.861	0.904	0.882	DT _{TF-IDF}	0.681	0.092	0.454	0.832	0.587
DT _{BERT}	0.756	0.284	0.797	0.737	0.766	DT _{BERT}	0.577	0.284	0.437	0.606	0.508
LR _{TF-IDF}	0.932	0.123	0.986	0.889	0.935	LR _{TF-IDF}	0.772	0.123	0.666	0.845	0.745
LR _{BERT}	0.904	0.109	0.916	0.894	0.905	LR _{BERT}	0.582	0.109	0.273	0.715	0.395
SGD _{TF-IDF}	0.940	0.047	0.928	0.951	0.939	SGD _{TF-IDF}	0.916	0.047	0.880	0.949	0.913
SGD _{BERT}	0.907	0.075	0.889	0.922	0.905	SGD _{BERT}	0.572	0.075	0.220	0.745	0.340
RF _{TF-IDF}	0.901	0.175	0.978	0.848	0.908	RF _{TF-IDF}	0.613	0.175	0.401	0.696	0.509
RF _{BERT}	0.861	0.223	0.944	0.809	0.871	RF _{BERT}	0.607	0.223	0.437	0.662	0.527
SVM _{TF-IDF}	0.925	0.134	0.983	0.880	0.929	SVM _{TF-IDF}	0.702	0.134	0.538	0.801	0.643
SVM _{BERT}	0.903	0.114	0.919	0.889	0.904	SVM _{BERT}	0.563	0.114	0.240	0.677	0.354
BERT	0.910	0.091	0.904	0.896	0.909	BERT	0.946	0.080	0.929	0.985	0.952
Sabiá-3	0.560	0.005	0.990	0.530	0.700	Sabiá-3	0.570	0.008	0.990	0.540	0.700
RAG _{EX}	0.540	0.013	0.990	0.520	0.680	RAG _{EX}	0.540	0.019	0.980	0.520	0.690
RAG _{IN_EX}	0.570	0.008	0.990	0.540	0.700	RAG _{IN_EX}	0.570	0.008	0.990	0.540	0.700

which present results for when the training set consists of LLM-generated fake news, while the test set contains only original fake news. Once again, the generative LLM-based approaches outperformed traditional ML methods and BERT-based one, reinforcing the hypothesis that they are more robust to shifts in data distribution and stylistic differences between original and synthetic fake news. Nevertheless, as previously discussed, we cannot rule out the possibility of data leakage from the LLM's training process influencing these results. Another interesting point is that, while all other methods experienced a significant drop in performance compared to experiments using only original fake news, the LLM maintained similar results. Additionally, it is important to highlight that RAG once again outperformed the direct use of the LLM in Fake.br dataset, reinforcing the relevance of the use

of external knowledge in this context.

The results also indicate that training on LLM-generated fake news alone is not sufficient for traditional ML models to generalize effectively to real-world fake news. This highlights the importance of maintaining human-labeled datasets in the development of fake news detection methods, as models trained solely on synthetic data may struggle to perform well on real-world misleading content. These trends change when both the training and test sets consist exclusively of LLM-generated fake news, as shown in Tables 3(d) and 4(d). In this configuration, traditional ML methods performed better, reinforcing the idea that the RAG approach excels only when there is a significant difference between the news in the training and test sets. When traditional models were trained on synthetic fake news, a simple text representation

approach like TF-IDF was enough to distinguish between classes, even outperforming BERT embeddings as vector representations. This aligns with the linguistic feature analysis presented in Table 2, where we previously observed significant differences in certain linguistic characteristics between true news and LLM-generated fake news. Although TF-IDF does not directly consider class labels, it can highlight words that are statistically more frequent or distinctive within subsets of documents, which may correlate with different categories.

As we observed in the previous experiments, the results from BERT and traditional ML methods were often lower when the training and test sets were composed of different categories of fake news. To investigate whether using a hybrid training set might improve performance, Tables 3(e) and 4(e) show the results when the training set consists of 50% original fake news and 50% LLM-generated fake news, while the test set contains only original fake news. On the other hand, Tables 3(f) and 4(f) display the results when the training set is hybrid, but the test set contains only LLM-generated fake news. These experiments were conducted to assess whether a mix of original and synthetic fake news in the training set could offer a better approach for detecting either human or synthetic fake news.

The results indicate that this hybrid approach performs better for traditional ML methods than training with only human-generated fake news to classify LLM-generated fake news (as shown in Tables 3(b) and 4(b)), or vice versa (Tables 3(c) and 4(c)). However, both configurations produce results significantly lower than when the training and test sets are of the same category (either human or synthetic fake news — Tables 3(a), 4(a), 3(d), and 4(d)). For instance, in the Fake.br experiments, SVM_{TF-IDF} in Table 3(c) obtained an F-measure that was 23% lower than in Table 3(a). This suggests that simply mixing original and synthetic fake news is not sufficient to allow the methods to generalize effectively. Moreover, this indicates that generating synthetic fake news from LLMs based on true news is not an optimal strategy for dataset augmentation. While generating synthetic fake news from human-generated fake news could yield better results, it is more challenging, as obtaining true news from reliable sources is relatively easier.

The only method that experienced a smaller drop in performance when using this hybrid approach was BERT. This suggests that, while the hybrid strategy may not be ideal for most methods, it could still be effective for BERT, which exhibited a less pronounced decline in performance compared to other models.

For the RAG-based approaches, the hybrid approach had no significant impact on performance, neither positive nor negative, even in the experiments with the original test set. Furthermore, in all experiments with the hybrid approach, RAG outperformed the LLM in terms of accuracy, precision, and F-measure on the Fake.br dataset. Since the LLM does not utilize the training data, this consistent superiority of RAG suggests that external knowledge plays a crucial role in achieving better results. On the FakeTrueBR dataset, however, the direct LLM approach and RAG performed nearly identically.

6 Conclusion

Fake news is a serious problem for society and a challenge for Natural Language Processing (NLP), especially in Portuguese, where resources are more limited. With the rise of LLMs, this type of content can now be generated quickly and convincingly, making detection even more complex. This study investigates how automatic fake news detection methods perform when faced with LLM-generated misleading content and if they can still perform well in this new scenario.

We conducted experiments using two Portuguese-language datasets with aligned news articles. For each true news article, we generated a fake version using an LLM model. Our experiments assessed the performance of various AI models, including traditional ML, a BERT-based model, LLM, and LLM+RAG. All models were asked to classify fake news written by humans versus that generated by LLMs. Overall, the results showed a performance decay when the models were trained on one type of fake news (human-written or LLM-generated) and tested on the other. We also observed that hybrid training, which includes both types of fake news, i.e., original and synthetic, improved performance but not significantly.

These results demonstrate that automatic classification approaches struggle to generalize across different types of fake news, highlighting a potential vulnerability in current detection systems. Hence, the development of more robust approaches, that can adapt to evolving tactics, is needed, especially as LLMs become more sophisticated.

Despite this general trend in the results, the classification strategies showed important differences. When the training and test datasets contained the same type of fake news (either human-written or LLM-generated), traditional ML classifiers, particularly SVM and SGD, along with the fine-tuned BERT, generally obtained the best results. However, when the data set contained mixed types, with LLM-generated fake news appearing only in the training or test set, generative LLM-based classifiers outperformed the other methods, especially the RAG approach that combined internal and external knowledge.

When a hybrid training set was used, containing both human-written and LLM-generated fake news, the fine-tuned BERT generally performed well. Although results were still lower compared to the experiments where the training and test sets matched, hybrid training proved to be a promising direction for improving generalization. In this work, we adopted a simple hybrid approach with an equal mix of 50% of both types of fake news. Future research could explore more sophisticated hybrid training strategies that might further enhance robustness against both types of fake news, including domain adaptation techniques, which may help the model to adjust to distribution shifts between train and test data.

Our evaluation considered only text-based classification, without incorporating multimodal features such as images or metadata, which are often present in real-world fake news. In future research, we intend to investigate how incorporating multimodal features, such as images, videos, and social context, can improve fake news detection in both human and LLM-generated fake news.

Furthermore, this work focused specifically on the Portuguese language, which has fewer NLP and ML resources compared to English. Future work could explore whether our findings can also be observed in other low-resource languages, such as Spanish, French, Hindi, and Indonesian, where similar challenges in fake news detection may arise due to limited labeled datasets and language-specific features.

Declarations

Authors' Contributions

Renato Moraes Silva: Writing - review & editing, Project administration, Formal analysis, Conceptualization, Methodology, Data curation. **Lucca Baptista Silva Ferraz:** Writing - original draft, Validation, Formal analysis, Data curation. **Fabio Kauê Araujo da Silva:** Writing - original draft, Validation, Formal analysis, Data curation. **Hazem Amamou:** Writing - review & editing, Validation, Methodology, Formal analysis. **Anderson Raymundo Avila:** Writing - review & editing, Validation, Methodology, Formal analysis, Conceptualization.

Competing interests

The authors declare that they have no competing interests.

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Availability of data and materials

The datasets generated and analysed during the current study are available in <https://github.com/renatosvmor/fake-news-llm-ptbr>.

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